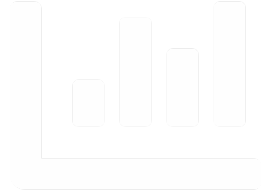


Topic 0



Lecture 0 · Introduction

National University of Singapore (NUS)

TODAY'S OBJECTIVES

1

Course Logistics

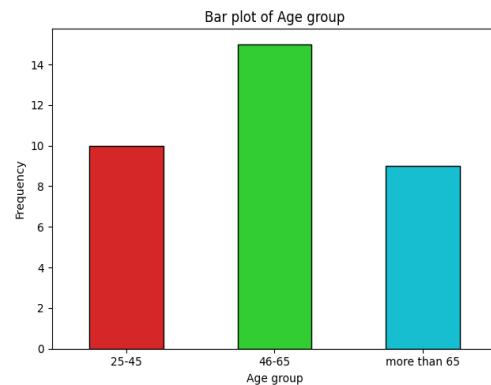
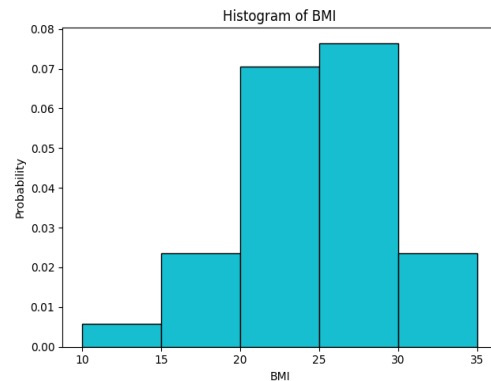
2

Statistics



Goals

- For users or consumers of statistics
- You will learn:
 - Simple distributions
 - Basic probability
 - Statistical concepts
- You will learn how to:
 - Produce and use statistical graphics



Goals

- You will learn how to:
 - Determine an appropriate statistical model for given datasets
 - Conduct sensible statistical inference
 - Understand statistical portions in reports or journals
 - Analyze real-world datasets using statistical software
 - Solve statistical questions about datasets
- You will gain experience in
 - Formulating statistical questions
 - Answering these statistical questions
 - Communicating findings to non-technical audience

How we achieve these goals

- Keep formulas and mathematics foundation to minimum
 - Focus on how to use the theorem in practical problems
- Rely on software for computation
- Focus on:
 - Understanding logic behind statistical decisions
 - Interpretation of software output

Materials

- Canvas:
 - Lecture notes
 - Tutorial questions and solutions
 - Datasets
 - Announcements
- References (**optional**):
 - Statistics: The Art and Science of Learning from Data, 5th Edition
 - Authors: Agresti, Franklin and Klingenberg
 - Hardcopy available for purchase from Co-op, or for borrowing from library

Lectures and Tutorials

- Lectures:
 - Pre-recorded lectures
 - Live lectures (recorded and uploaded on Canvas)
 - Finish all previous lectures before attending live lectures
- Tutorials:
 - Register online
 - Help you with tutorial questions
 - Support you in learning software
 - All conducted physically

Assessments

- Requires use of statistical software
- Contains software output
- You will be tested on:
 - How to produce output using software
 - Writing code to obtain numerical or graphical results

How you can get help

- Need help to better understand notes/tutorials, or write code using software
- Check the FAQ folder
- Post questions on the Discussion Forum on Canvas
- Email questions to lecturer or teaching assistants
 - Put "ST1131" in the title
- Email to book consultation
 - Zoom or face to face
 - One-to-one or group

TODAY'S OBJECTIVES

1

Introduction to R programming

2

Exploratory Data Analysis (EDA)

3

Collecting Data

4

Probability

5

Random Variables (Discrete and Continuous)

6

Sampling Distribution

7

Statistical Estimation

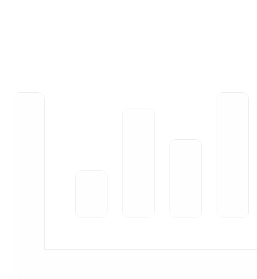
8

Hypothesis Testing

9

Linear Regression

- Limitations of linear regression (if time permits)



What is statistics?

Applications of Statistics

- Used in almost all domains
- Business:
 - Managers analyze results of marketing studies about new products to predict sales
 - Measure employee performance
- Finance: Study stock returns and investment opportunities
- Medicine:
 - Evaluate effectiveness of new drugs
 - Identify the main cause of fatal diseases
- News, advertisements, surveys, political campaigns, etc.

Statistics

- Answer a question
- Make a decision
- Needs information gathered from experiments or surveys
 - Information gathered is collectively called data

Statistics



DEFINITION

Statistics is the art and science of designing studies and analysing data that those studies produce. Its ultimate goal is to translate data into knowledge, that allows us to make informed and objective decisions.

Investigating using statistics

STEP-BY-STEP

1

Form a statistical question.

- Does smoking lead to higher chance of having lung cancer?
- At our level, is usually given

2

Design

- How we collect data

3

Description

- Analyze data collected

4

Inference

- Interpret results

Example 1: Price of Resale Flats

- In Singapore, HDB is a type of flat built and sold by the government to Singaporeans and Permanent Residents only
- About 80% of the population live in HDB flats
- Flats could be brand new or resale flats
- We are interested in predicting the price of HDB resale flats

Example 1: Price of Resale Flats

STEP-BY-STEP

1

Form a statistical question.

- What are the factors that affect the price of a resale flat?

2

~~Design~~ Data provided

- Retrieve data from: <https://data.gov.sg/dataset/resale-flat-prices>

3

Description

4

Inference

Example 2: Aspirin and Heart Disease

- Relationship between taking aspirin and heart disease
- Heart disease is the most common cause of death in industrialized nations
- Does regular aspirin intake reduce deaths from heart attacks?

Example 2: Aspirin and Heart Disease

- Harvard Medical School survey

Took aspirin regularly

0.9%

Did not take aspirin

1.7%

Example 2: Aspirin and Heart Disease

STEP-BY-STEP

1

Form a statistical question.

- Does regular aspirin intake reduce deaths from heart attacks?

2

Design

- Who to include in the study?
- How many people to join the study?
- How to assign the aspirin and the placebo?
- How long do you follow them? 5 years? 10 years?
- ...

3

Description

- Percentage of individuals who had heart disease in Aspirin group VS Placebo group

4

Inference

- Whether aspirin reduces the chances of having heart disease

Population



DEFINITION

Population is the total set of all subjects in which we are interested.

Sample



DEFINITION

Sample is the subset of the population for whom we have or plan to have data for.
This subset is often randomly selected.

Using sample to understand population

- Population is too big to get exact value
- Have a representative of the population which is a random sample selected from the population
- Representative should have any information in the population

Example 1: Price of Resale Flats

Population

All HDB resale flats in Singapore

Sample

All flats included in the data
provided from the website

Example 2: Aspirin and Heart Disease

Population

US people in the age range 35-65

Sample

All US people involved in the study

- Aspirin
- Placebo
- Age range 35-65

Parameter and Statistic



DEFINITION

Parameter is a numerical summary of the population. It is unknown.

KNOWN AFTER SOME EVENT

- Proportion voting for a party is unknown before election ends

UNKNOWN FOREVER

- Exact proportion of Singaporeans that have diabetes

Parameter and Statistic



DEFINITION

Statistic is a summary of a sample taken from the population.

We compute it based on the given sample.

We use statistic to estimate the parameter.

NOTATION

p = population proportion

$\hat{p}$ = sample proportion

EXAMPLE

- Estimate the proportion of Singaporeans with diabetes
- Use $\hat{p}$ to estimate p

Parameter and Statistic



DEFINITION

Statistic is a summary of a sample taken from the population.

We compute it based on the given sample.

We use statistic to estimate the parameter.

NOTATION

μ = population mean height

$\bar{X}$ = sample mean height

EXAMPLE

- Estimate the mean height of all NUS students
- Use $\bar{X}$ to estimate μ
- Take into account the randomness of the sample
- Sample is good representative of population

Random Sampling

- Every subject in the population has an equal chance of being selected in the sample
- Allows powerful inference about population
- Crucial for performing experiments
- Value of statistic changes, but parameter does not change